EXECUTIVE SUMMARY

FACTORY REALLOCATION & SHIPPING OPTIMIZATION RECOMMENDATION SYSTEM FOR NASSAU CANDY DISTRIBUTOR

1. Project Overview

The Factory Reallocation & Shipping Optimization Recommendation System is a data-driven decision-support project developed to evaluate the existing factory assignments and shipping performance of Nassau Candy Distributor. The project aims to identify potential opportunities where assigning a product to an alternative factory may improve predicted shipping lead time while maintaining reasonable financial stability.

The system combines data analytics, exploratory data analysis, route-level performance analysis, machine learning, clustering, what-if scenario simulation, and recommendation scoring. The overall objective is to move from historical performance reporting toward predictive and scenario-based decision support.

2. Business Problem

Nassau Candy Distributor operates with fixed product-to-factory assignments. Although this provides a stable manufacturing structure, it may not always result in the most efficient shipping outcomes for every destination region and shipping condition.

Differences in factory location and operational patterns can contribute to variations in shipping lead time across products and regions. A systematic approach is therefore required to evaluate whether alternative factory assignments could provide measurable improvements.

The project addresses this problem by predicting shipping lead time, simulating alternative factory assignments, comparing the predicted outcomes, and generating structured recommendations for potential reallocation opportunities.

3. Dataset and Scope

The analysis was performed using a transaction-level Nassau Candy Distributor dataset containing 10,194 records and 18 original fields. The dataset includes order details, customer information, product information, shipping mode, geographical attributes, sales, units, gross profit, cost, order date, and ship date.

The project covers:

Dataset Component	Count
Transaction Records	10,194
Unique Orders	8,549
Unique Products	15
Unique Customers	5,044
Factories	5
Destination Regions	4
Shipping Modes	4
Countries	2

The product-to-factory mapping supplied for the project was incorporated into the dataset, allowing the current factory configuration to be treated as the baseline for scenario comparison.

4. Data Preparation and Analysis

The raw dataset was cleaned and validated before analysis. Column names were standardized, date fields were converted into valid datetime values, and missing-value and duplicate checks were performed.

No missing values were identified across the dataset, and no completely duplicated transaction records were found. Repeated Order IDs, Customer IDs,

and Product IDs were retained because the dataset represents transaction-level business records.

Several derived features were created, including Lead_Time_Days, Profit_Margin, Cost_Percentage, Sales_Per_Unit, Cost_Per_Unit, Profit_Per_Unit, Order_Year, Order_Month, Order_Quarter, Order_DayOfWeek, Is_Weekend, and Current_Factory.

Lead time was calculated as the difference between shipment date and order date. The resulting average lead time was 1,320.84 days, with a median of 1,274 days.

5. Exploratory and Route-Level Findings

Exploratory analysis was used to understand product, regional, factory, shipping-mode, and financial patterns.

Standard Class was the dominant shipping mode, representing approximately 60.04% of records. The Pacific region had the highest transaction volume among the four destination regions.

The Chocolate division accounted for approximately 96.57% of all records, while the remaining records belonged to the Sugar and Other divisions.

For more detailed operational analysis, transactions were aggregated into 20 observed factory-region route combinations. Route-level analysis considered average and median lead time, lead-time variability, sales, gross profit, profit margin, and other performance indicators.

This analysis demonstrated that shipping performance is not uniform across all factory-region combinations and provided the foundation for route clustering and scenario analysis.

6. Route Clustering

K-Means clustering was applied to the route-level performance data to group factory-region combinations with similar operational and financial characteristics.

The clustering process considered Average Lead Time, Lead-Time Variability, Average Profit Margin, Profit Per Record, and Sales Per Record.

Cluster values from 2 to 6 were evaluated using inertia and silhouette score. Five clusters were selected because they produced the highest silhouette score among the tested configurations, with a score of approximately 0.588.

The resulting clusters represent groups of routes with similar characteristics and are used for analytical interpretation rather than direct performance ranking.

7. Predictive Modeling

A machine learning approach was developed to predict shipping lead time using product, factory, region, shipping mode, division, country, temporal, sales, units, gross profit, cost, and profitability-related features.

The dataset was divided into 80% training data and 20% testing data using a fixed random state of 42.

Three regression models were evaluated:

Linear Regression achieved an MAE of 181.39 days, RMSE of 182.51 days, and R^2 score of 0.5290.

Random Forest Regressor achieved an MAE of 148.77 days, RMSE of 177.17 days, and R^2 score of 0.5562.

Gradient Boosting Regressor achieved an MAE of 159.87 days, RMSE of 169.59 days, and R^2 score of 0.5933.

Gradient Boosting Regressor was selected because it achieved the highest R^2 score and the lowest RMSE among the evaluated models.

Feature-importance analysis showed a strong contribution from temporal variables, particularly Order Year and Order DayOfWeek. This is an important interpretation point because the model learns patterns present in the available historical data.

Regression Model	MAE (Days)	RMSE (Days)	R^2 Score
Linear Regression	181.39	182.51	0.5290
Random Forest Regressor	148.77	177.17	0.5562
Gradient Boosting Regressor	**159.87**	**169.59**	**0.5933**

8. Factory Reallocation Scenario Analysis

The selected Gradient Boosting model was used to perform what-if scenario simulation across the five available factories.

Each transaction was evaluated under alternative factory assignments, resulting in 50,970 transaction-level factory scenarios.

The scenarios compared the predicted lead time of a candidate factory with the predicted lead time under the current factory assignment. Lead-time reduction was calculated both in days and percentage terms.

At the product-region level, 265 product-region-candidate factory combinations were evaluated. After removing the current factory and scenarios with zero or negative predicted improvements, 33 positive reallocation opportunities were identified across 53 observed product-region combinations.

This represents recommendation coverage of approximately 62.26%.

9. Recommendation and Optimization Framework

The recommendation engine evaluates potential reallocation scenarios using multiple indicators rather than relying only on shipping speed.

The main indicators include Speed Score, Scenario Confidence Score, Profit Stability Score, Optimization Score, and Final Risk Level.

The optimization framework gives 70% priority to shipping-speed improvement and 30% priority to profit stability.

The Scenario Confidence Score acts as a project-specific confidence proxy based on transaction volume and model performance. It is not a formal statistical confidence interval.

The Profit Stability Score is based on available historical profit-related indicators and does not represent a direct prediction of future profit after factory reassignment.

10. Key Results

The 33 identified positive reallocation opportunities produced an average predicted lead-time reduction of approximately 7.07 days, with a median reduction of 5.44 days.

The average relative improvement across these opportunities was approximately 0.56%.

The risk distribution of the recommended opportunities was:

Risk Level	Opportunities	Percentage
High	17	51.52%
Medium	4	12.12%
Low	12	36.36%
Total	**33**	**100%**

Risk Distribution table

At the overall system level, the model-based optimized scenario reduced weighted predicted lead time from approximately 1,320.99 days to 1,315.81 days.

This represents an estimated overall reduction of approximately 5.18 days, or 0.39%.

These values are what-if predictions generated by the model and should not be interpreted as guaranteed operational improvements.

Key Performance Indicator	Result
Product-Region Combinations Evaluated	53
Positive Reallocation Opportunities	33

Recommendation Coverage	62.26%
Average Predicted Lead-Time Reduction	7.07 days
Median Predicted Lead-Time Reduction	5.44 days
Average Relative Improvement	0.56%
Overall Predicted Lead-Time Reduction	5.18 days
Overall Predicted Improvement	0.39%

Key Performance Indicator

11. Business Recommendations

The analysis suggests that the identified positive reallocation opportunities can be considered for further operational evaluation.

Higher-impact scenarios with larger predicted lead-time reductions can be prioritized for detailed validation. At the same time, recommendation risk should be considered so that scenarios with lower confidence are not treated in the same way as stronger opportunities.

Before implementing any factory reassignment, the shortlisted recommendations should be validated against practical factors such as factory capacity, inventory availability, production capabilities, transportation costs, manufacturing constraints, and service-level requirements.

The dashboard can be used as a scenario-planning tool to compare alternative factory configurations and support structured decision-making.

12. Dashboard and Decision Support

A Streamlit-based interactive dashboard was developed to present the analytical findings and scenario results in a user-friendly format.

The dashboard provides sections for executive overview, factory recommendations, scenario simulation, route performance, and predictive model evaluation.

Users can select products, destination regions, shipping modes, and candidate factories to examine model-based predicted lead times and compare alternative factory scenarios.

The dashboard provides a practical interface for exploring the project's analytical outputs without requiring direct interaction with the underlying Python code.

13. Business Value

The project provides a structured framework for moving from descriptive business analysis toward predictive and scenario-based decision support.

Instead of only identifying historical shipping patterns, the system evaluates alternative factory configurations and quantifies their predicted operational impact.

The inclusion of profit-stability indicators also ensures that operational improvement is considered alongside financial characteristics.

The developed framework can support planners and decision-makers in screening potential factory reallocation scenarios before conducting detailed operational feasibility analysis.

14. Limitations

The project has several limitations that should be considered when interpreting the results.

The dataset does not contain historical observations of every product being manufactured at every possible factory. Consequently, alternative factory results are model-based what-if estimates rather than directly observed outcomes.

The available dataset does not provide factory-specific transportation costs, production costs, inventory constraints, or factory capacity information. Therefore, direct monetary savings from factory reallocation cannot be calculated.

The observed lead-time values are derived directly from the supplied order and shipment dates and were not artificially adjusted.

The predictive model also contains estimation error, and its strong dependence on temporal variables should be considered when interpreting the scenario results.

15. Future Scope

The system can be further enhanced by incorporating transportation distance, actual logistics costs, factory capacity, inventory availability, production constraints, and product-specific manufacturing capabilities.

Advanced optimization techniques such as linear programming or mixed-integer optimization could also be incorporated to evaluate multiple real-world constraints simultaneously.

The dashboard can be connected to regularly updated operational data to support continuous monitoring and dynamic recommendation generation.

With richer operational and financial information, the current predictive what-if framework could be extended into a more comprehensive factory and logistics optimization system.

16. Conclusion

The Factory Reallocation & Shipping Optimization Recommendation System successfully combines data analytics, machine learning, route analysis, clustering, and what-if simulation to create a structured decision-support framework for Nassau Candy Distributor.

The project analyzed 10,194 transaction records, evaluated multiple predictive models, selected Gradient Boosting Regressor for scenario simulation, and identified 33 positive product-region reallocation opportunities.

The model-based scenario analysis indicated an average predicted improvement of 7.07 days across the identified opportunities and an overall system-level predicted improvement of approximately 5.18 days.

The project demonstrates how historical transaction data can be transformed into practical predictive insights while maintaining transparency about model limitations and uncertainty.

The final system should be used as a decision-support tool, with recommended scenarios validated against actual operational, financial, and capacity constraints before implementation.

Management Takeaway

The analysis identified 33 potential factory reallocation opportunities across the observed product-region combinations. While the model indicates potential shipping-time improvements, these scenarios should be validated against factory capacity, transportation cost, inventory availability, and production constraints before implementation.